

EXPERIMENT:

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C PROGRAMS

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Common syntax in C for all C programs:

Output:

#include <stdio.h>

int main() { # int main() is standard declaration

Hello, World!

printf("Hello, World!"); # main() returns value to int
return 0;

}

C prgm for relational operators:

Output:

#include <stdio.h>

int main() {

a is not equal to b

int a = 20;

a is greater than b

int b = 10;

a is not less than b

int c;

if (a == b) {

printf("a is equal to b\n");

}

else

printf("a is not equal to b\n");

if (a > b) {

printf("a is greater than b\n");

}

else

printf("a is not greater than b\n");

if (a < b) {

printf("a is less than b\n");

}

else

printf("a is not less than b\n");

}

C prgm for assignment operators:

```
#include <stdio.h>
```

```
int main() {
```

```
    int a = 20;
```

```
    int c;
```

```
    c = a;
```

```
    printf("Value of c = %d \n", c);
```

```
    c += a;
```

```
    printf("Value of c += %d \n", c);
```

```
    c -= a;
```

```
    printf("Value of c -= %d \n", c);
```

```
    c *= a;
```

```
    printf("Value of c *= %d \n", c);
```

```
    c /= a;
```

```
    printf("Value of c /= %d \n", c);
```

```
    c %= a;
```

```
    printf("Value of c %= %d \n", c);
```

```
    c <<= a;
```

```
    printf("Value of c <<= %d \n", c);
```

```
    c >>= a;
```

```
    printf("Value of c >>= %d \n", c);
```

```
    return 0;
```

```
}
```

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C prgm for & (memory address operator):

```
#include <stdio.h>
```

```
int main() {
```

```
    int age = 23;
```

```
    printf("%d \n", age);
```

```
    printf("%d", &age);
```

```
}
```

Output:

Value of c = 20

Value of c = 40

Value of c = 20

Value of c = 400

Value of c = 20

Value of c = 0

Value of c = 0

Value of c = 0

Output:

23

2052321260

C prgm to find avg of 3 subj of students :

Output :

```
#include <stdio.h>
int main() {
    int a;
    int b;
    int c;
    char name[5];
    float result;
    printf("\n Pls enter name:");
    scanf("%s", &name);
    printf("\n Pls enter mark1:");
    scanf("%s", &a);
    printf("\n Pls enter mark2:");
    scanf("%s", &b);
    printf("\n Pls enter mark3:");
    scanf("%s", &c);
    result = (a+b+c)/3;
    printf("Average of marks = %f", result);
    return 0;
}
```

Pls enter name : Sara

Pls enter mark1 : 20

Pls enter mark2 : 30

Pls enter mark3 : 40

Average of marks = 30.000000

22-1-26 # C prgm to calculate area of circle :

Output :

```
#include <stdio.h>
int main() {
    float radius, area;
    printf("Enter radius:");
    scanf("%f", &radius);
    area = 3.14 * radius * radius;
    printf("area of circle = %f", area);
}
```

Enter radius : 2

area of circle : 12.560000

C prgm to print prime or not.

include <stdio.h>

int main(){

int i, n, count=0;

printf("Enter the number:");

scanf("%d", &n);

for (i=1; i<=n; i++){

if (n%i==0){

count++;

}

if (count==2){

printf("number is prime");

}

else {

printf("number is not prime");

}

}

Output:

Enter the number: 3

number is prime

Enter the number: 6

number is not prime.

C prgm to display no. in ascending/descending order.

include <stdio.h>

int main(){ int i=1;

while (i<10){

printf("%d", i);

i++; //i-- for descending.

}

}

Output:

1 2 3 4 5 6 7 8 9

C prgm to calculate area of rectangle.

```
#include <stdio.h>
```

```
void main() {
```

```
    int l, b, a;
```

```
    printf("enter len of rectgl");
```

```
    scanf("%d", &l);
```

```
    printf("enter brdth of rectgl");
```

```
    scanf("%d", &b);
```

```
    a = l * b;
```

```
    printf("area of rectgl: %d", a);
```

```
}
```

Output:

Enter len of rectgl: 2

Enter brdth of rectgl: 4

area of rectgl: 8

C prgm to calculate area of square.

```
#include <stdio.h>
```

```
void main() {
```

```
    int s, a;
```

```
    printf("Enter len of sq:");
```

```
    scanf("%d", &s);
```

```
    a = s * s;
```

```
    printf("Area of sq: %d", a);
```

```
}
```

output:

Enter len of sq: 2

Area of sq: 4

* Call by Function:

```
#include <stdio.h>
```

```
void myfunction() {
```

```
    printf("Hello");
```

```
}
```

```
int main() {
```

```
    myfunction();
```

```
}
```

Output:

Hello.

C prgm to calculate sum of nos.

Output:

```
#include <stdio.h>
void clts() {
    int a=5;
    int b=10;
    int sum;
    sum = a+b;
    printf("Sum of given nos. is: %d", sum);
}
int main() {
    clts();
}
```

Sum of given nos is : 15.

C Function Parameters

Output:

```
#include <stdio.h>
void myfrnds(char name[]) {
    printf("Hii, %c", name); or // %s
}
int main() {
    myfrnds("Priyanka\n");
    myfrnds("Sneha\n");
}
```

Hii, Priyanka
Hii, Sneha.

C prgm to display reverse of given nos

Output:

```
#include <stdio.h>
void myfrnds(char name[]) {
    printf("Hii, %c", name); // %s
}
int main() {
    myfrnds("Priyanka\n");
    myfrnds("Sukriti\n");
    myfrnds("Deepti\n");
}
```

Hii, Priyanka
Hii, Sukriti
Hii, Deepti

C prgm to find factorial of number.

Output :

#include <stdio.h>

Enter number : 5

void main() {

factorial of no: 120

int i, fact = 1;

printf("Enter number:");

scanf("%d", &i);

while (i > 0) {

fact = fact * i;

i--;

}

printf("\n factorial of no: %d, fact);

}

C prgm for break & continue.

Output :

for continue
output
would be

#include <stdio.h>

int main() {

int i;

for (i = 0; i < 10; i++) {

if (i == 4) {

break; // continue

}

printf("%d\n", i);

}

return 0;

}

C prgm to access an element from Array :

Output :

#include <stdio.h>

20

int main() {

int myNum[] = {20, 30, 40, 50};

printf("%d\n", myNum[0]);

}

C prgm to get Array sizes in terms of bytes:

#include <stdio.h>

int main() {

int myNum[] = {20, 30, 40, 50};

printf("%d\n", size of (myNum));

}

Output:

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2-D Array in C prgm

#include <stdio.h>

int main() {

int matrix[2][3] = {{1, 4, 5}, {3, 6, 2}};

printf("%d", matrix[0][2]);

}

Output:

5

	(0) 0	(0) 1	(0) 2
Row 0	1	4	5
Row 1	3	6	2

C prgm to create & access string:-

#include <stdio.h>

int main() {

char greet[] = "Hello World";

printf("%c", greet[2]);

}

Output:-

%c to print only 1 character

— x —

include <stdio.h>

int main() {

int i;

char greet[] = "Hello World";

for (i = 0; i < 11; i++) {

printf("%s\n", greet[i]);

}

return 0;

}

Output:

Hello World

%s to print whole string.